

UNIT 4: LIFE ISSUES

Chapter 29: Ecodriving

COMPREHENSIVE ALL-IN-ONE REVISION BOOKLET

Course Theory • Bilingual Lexicon • 40 Q&A Active Recall • Exam Evaluation Quiz

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PRODUCTION SPECIFICATIONS

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Target Exam: Tunisian National Baccalaureate (English Subject)

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Ecodriving is an innovative and highly effective driving behavior designed to improve fuel efficiency, reduce greenhouse gas emissions, and promote sustainable road safety. As outlined in the Tunisian Baccalaureate curriculum (Unit 4: Life Issues), ecodriving transitions our approach to transportation from passive consumption to active environmental stewardship. By altering simple behind-the-wheel habits, drivers can substantially reduce their **carbon footprint** and help mitigate global ecological concerns.

Core Principles of Eco-Driving

1. **Smooth Acceleration:** Avoid aggressive acceleration ('jackrabbit starts'). Gently increasing speed optimizes fuel combustion.
2. **Early Shifting:** Manual drivers should shift gears up as early as possible (around 2,000 RPM) to keep the engine load efficient.
3. **Maintain Constant Speed:** Frequent braking and accelerating squanders kinetic energy. Using cruise control on highways helps maintain stable momentum.
4. **Anticipate Traffic:** Look far ahead. Letting the car coast down with the engine brake instead of braking abruptly prevents friction energy loss.
5. **Limit Excess Idling:** Modern engines require no warm-up. If stopped for more than 30 seconds, turning off the engine saves fuel and reduces localized pollution.
6. **Vehicle Maintenance & Aerodynamics:** Under-inflated tires increase rolling resistance by 3%, while roof racks drastically increase aerodynamic drag.

Fuel Efficiency Improvement Potential (%) by Eco-Driving Tech

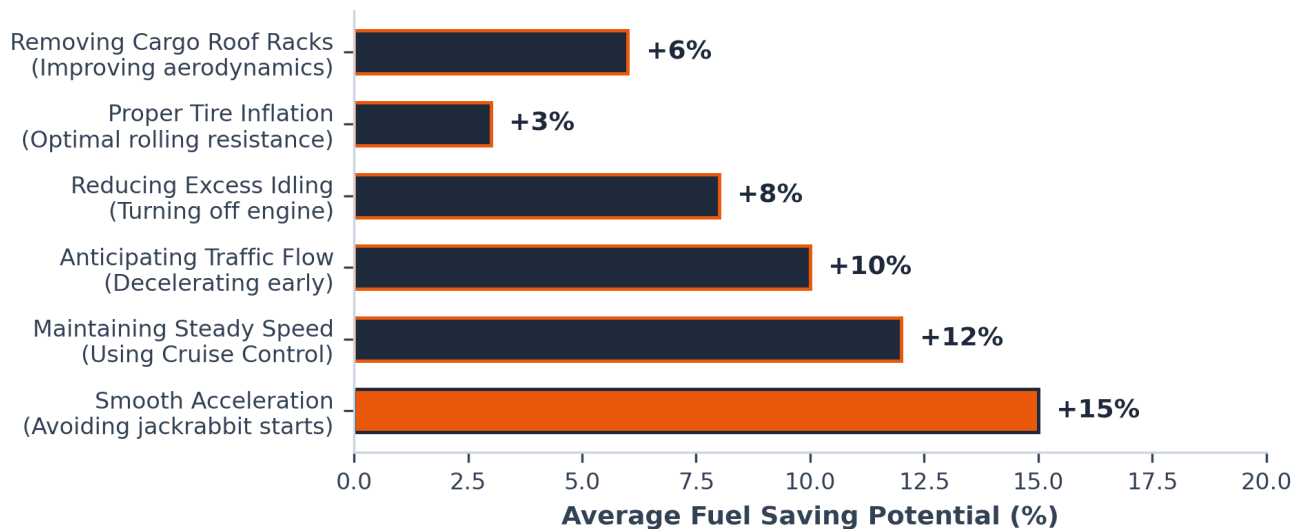


Figure 1.1: Average percentage of fuel saved through various ecodriving practices (mathbacai.com database)

Thematic Vocabulary Portfolio (Essential BAC Terms)

Baccalauréat Term	Phonetics (API)	Definition / Context	French Translation
ecodriving	/ˈiː.kəʊˌdraɪ.vɪŋ/	An eco-friendly style of driving that reduces fuel use.	éco-conduite
fuel efficiency	/ˈfjuː.əl ɪˈfɪʃ.ən.si/	The level of distance traveled relative to fuel consumed.	efficacité énergétique
carbon footprint	/ˈkɑː.bən ˈfʊt.prɪnt/	The total amount of greenhouse gases caused by an action.	empreinte carbone
aerodynamic drag	/,eə.rəʊ.darˈnæm.ɪk dræg/	Resistance forces acting on a vehicle moving through air.	traînée aérodynamique
excess idling	/ɪkˈses ˈaɪ.dl.ɪŋ/	Leaving a vehicle's engine running while stationary.	ralenti excessif
tire inflation	/ˈtaɪər ɪnˈfleɪ.ʃən/	The process of filling tires to their optimal pressure.	gonflage des pneus
to mitigate	/tuː ˈmɪt.ɪ.ɡeɪt/	To make something bad or harmful less severe or serious.	atténuer / réduire
acceleration	/əkˌsel.əˈreɪ.ʃən/	An increase in speed or the rate of speed increase.	accélération

Grammar Focus: Expressing Advice, Suggestions & Regret

In the context of ecodriving and environmental protection, expressing advice (présenter un conseil) and regret (regretter des actions passées) are highly tested BAC competencies.

1. Present / Future Advice:

To give advice, we use the following modals followed by a **Bare Infinitive**:

- Should / Shouldn't: You **should check** your tire pressure monthly.
- Ought to / Ought not to: We **ought to reduce** excess idling to curb emissions.
- Had better / Had better not: You **had better not accelerate** aggressively if you want to save money.

2. Past Regret and Criticism:

To criticize past environmental negligence or express regret, we use:

- **Should have + Past Participle (V3)**: We **should have adopted** eco-driving habits decades ago. (Regret: we didn't do it).
- **Shouldn't have + Past Participle (V3)**: He **shouldn't have left** his car idling for twenty minutes. (Criticism: he did leave it running).

Phonetics Module: Compound Noun Stress & Silent Letters

1. Stress in Compound Words: In compound nouns related to driving and ecology, the primary word stress shifts systematically to the **first element**. This is crucial for speaking and stress-matching BAC questions:

- **ECO-driving** (not eco-DRIVING)
- **FOOTprint** (not foot-PRINT)
- **CRUISE control** (not cruise-CONTROL)

2. Silent Letters Discrimination: Watch out for silent letters in examination vocabulary:

- **'h' in exhaust** /ɪg'zɔ:st/ → The 'h' is completely silent.
- **'g' in alignment** /ə'lain.mənt/ → The 'g' is silent.
- **'t' in soften** /'sɒf.ən/ → The 't' is silent.

Bac Model Writing: Argumentative Essay on Eco-Driving

Topic: Some people argue that saving fuel is an individual driver's choice, while others think that governments must enforce eco-driving regulations.

Model Response (12 Lines):

Although some citizens assert that driving styles belong strictly to personal freedom, global ecological degradation requires collective and governmental intervention. First and foremost, individual awareness remains fragile; consequently, many motorists continue to indulge in aggressive acceleration and excess idling. Therefore, authorities **should enforce** strict eco-driving instruction during license training and penalize high-emission vehicles. In doing so, we not only decrease fossil fuel dependency but we also **mitigate** public health hazards from tailpipe pollution. Ultimately, protecting our common atmospheric envelope must not depend on mere volunteerism; instead, it deserves structural regulation.

PART 2: THE ACTIVE RECALL REPOSITORY (40 Q&A;)

Ce recueil de **40 questions-réponses clés** est formaté de manière rigoureuse pour l'auto-évaluation. Pliez la page sur la colonne de gauche (Questions) pour vous tester de manière active, puis vérifiez vos réponses à droite.

Q1: What is the basic definition of ecodriving?	A1: An eco-friendly, systematic style of driving that minimizes fuel consumption, greenhouse gas emissions, and wear-and-tear on vehicles.
Q2: What does 'carbon footprint' represent in Unit 4?	A2: The total volume of carbon dioxide and other greenhouse gases emitted into the atmosphere due to specific human transport and daily choices.
Q3: Explain the term 'aerodynamic drag' in driving context.	A3: The air resistance force that pushes against a vehicle as it travels. High speeds and roof racks dramatically increase this drag, wasting fuel.
Q4: What does the verb 'to mitigate' mean?	A4: To alleviate, decrease, or make less intense or severe. Ecodriving is key to mitigating environmental decay.
Q5: How do we translate 'disposable income' to French?	A5: Le revenu disponible (partie du revenu dont dispose un ménage après impôts et cotisations).
Q6: What is 'excess idling' and why is it dangerous?	A6: Leaving a vehicle's engine running while stationary for over 30 seconds, which needlessly burns fuel and expels toxic emissions.
Q7: What is 'tire inflation' and how does it relate to eco-driving?	A7: The amount of air pressure inside vehicle tires. Proper tire inflation minimizes rolling resistance, boosting fuel mileage.
Q8: Translate 'greenhouse effect' to French.	A8: L'effet de serre (le réchauffement de l'atmosphère terrestre dû à l'accumulation de gaz polluants).
Q9: Define the term 'jackrabbit start' used in the text.	A9: Sudden, rapid, and aggressive acceleration from a standstill, which consumes immense quantities of fuel.
Q10: Translate 'fuel consumption' to French.	A10: La consommation de carburant.
Q11: What is the synonym of 'to squander' in this unit?	A11: To waste or throw away resources (money, time, fuel) recklessly.
Q12: What is the French equivalent of 'to customize'?	A12: Personnaliser ou adapter selon les préférences de l'utilisateur.
Q13: How do we express advice in the present in English?	A13: By using modals like 'should', 'ought to', or 'had better' followed directly by a bare infinitive verb.
Q14: Transform this advice: 'Check your tires' (using should).	A14: You should check your tires regularly to maintain fuel efficiency.
Q15: Transform this: 'Do not accelerate aggressively' (using had better).	A15: You had better not accelerate aggressively if you wish to conserve fuel.

Q16: What formula do we use to express past regret or criticism?	A16: Should have (or shouldn't have) + Past Participle (V3 form).
Q17: Translate: 'Tu aurais dû éteindre le moteur.'	A17: You should have turned off the engine while waiting in the car.
Q18: Translate: 'Il n'aurait pas dû acheter ce gros SUV polluant.'	A18: He shouldn't have bought that highly-polluting, heavy SUV.
Q19: What is the structural difference between 'should' and 'should have'?	A19: 'Should' takes a base verb for present advice, whereas 'should have' takes a past participle for past regret.
Q20: Complete: 'If we had driven slowly, we ___ (save) fuel.' (Type 3 Conditional)	A20: If we had driven slowly, we would have saved fuel.
Q21: Rewrite using unless: 'If you do not drive smoothly, you will waste fuel.'	A21: Unless you drive smoothly, you will waste fuel.
Q22: Can we write: 'You ought to checking tire pressure'?	A22: No, 'ought to' must be followed by a bare infinitive: 'You ought to check tire pressure'.
Q23: Can we use 'should not have + verb' for future advice?	A23: No. It is strictly used to criticize past events that have already occurred.
Q24: Complete: 'You had better ___ (not waste) electricity.'	A24: You had better not waste electricity.
Q25: Where is the primary stress in 'ecodriving'?	A25: On the first element: 'ECO-driving' (indicated by stress mark /'i:.kəʊ.drɑi.viŋ/).
Q26: Is the letter 'h' pronounced in 'exhaust'?	A26: No, the 'h' is silent. It is transcribed phonetically as /ɪg'zɔ:st/.
Q27: What is the silent letter in 'alignment'?	A27: The letter 'g' is completely silent (/ə'laɪn.mənt/).
Q28: Where is the primary stress in the word 'environment'?	A28: On the second syllable: en-VI-ron-ment (/ɪn'vaɪ.rən.mənt/).
Q29: Are 'footprint' and 'cruise control' stressed on the first or second word?	A29: On the first word: 'FOOTprint' and 'CRUISE control'.
Q30: Classify the past -ed ending of 'mitigated' (/ɪd/, /t/ or /d/)?	A30: It ends in /ɪd/ because the base verb 'mitigate' ends in a 't' sound.
Q31: Classify the past -ed ending of 'idled' (/ɪd/, /t/ or /d/)?	A31: It ends in /d/ because 'idle' ends in a voiced vowel sound.

Q32: What is the silent letter in 'debt'?	A32: The letter 'b' is completely silent (/det/).
Q33: What is Mrs. Cheryl Heron's main claim about school structure?	A33: She emphasized personalized paths based on ability rather than age to boost student confidence.
Q34: Name two main organizations supporting global literacy.	A34: UNESCO (United Nations Educational, Scientific and Cultural Organization) and UNICEF.
Q35: How does the School of the Air in Australia bypass geographical isolation?	A35: It uses radio transmissions and satellite technology to provide remote learning to outback pupils.
Q36: What was Ahmed Zewail's contribution to science?	A36: He invented femtosecond spectroscopy, allowing observation of chemical bonds breaking in real time.
Q37: What did James Hillier co-design in 1938?	A37: He co-designed the first functional high-resolution electron microscope in North America.
Q38: In the poem 'The Daffodils', what figure of speech is 'lonely as a cloud'?	A38: It is a simile, comparing the narrator's state of loneliness to a solitary cloud.
Q39: In 'Sonnet 18', what is the purpose of the poem?	A39: To immortalize the beauty of the poet's beloved through eternal verse, triumphing over physical decay.
Q40: What does Jaques mean by 'All the world's a stage' in As You Like It?	A40: He compares human life to a play, where everyone performs during their seven consecutive ages of life.

PART 3: EXAM EVALUATION QUIZ & SUMMARY MEMOS

Diagnostic Mock Exam (10 Questions)

This mock quiz is designed to simulate authentic BAC examination challenges. Test yourself, then review the detailed solutions below.

Section A: Multiple Choice (QCM)

- 1. Correct past regret expression:** We didn't inflated our tires. We wasted fuel.
A) We should have inflated our tires.
B) We ought to inflate our tires.
C) We shouldn't have inflated our tires.
- 2. Identify the word with the silent letter:**
A) Acceleration
B) Exhaust
C) Footprint
- 3. Complete: Unless you _____ your car, you will continue to consume too much fuel.**
A) don't align
B) align
C) wouldn't align
- 4. Where does the primary stress fall in the compound word 'ecodriving'?**
A) On the second element ('driving')
B) On the first element ('eco')
C) Stressed equally on both
- 5. Which past -ed ending is pronounced /ɪd/?**
A) idled
B) mitigated
C) aligned

Section B: True or False (with justification)

6. [True/False] An aerodynamic drag of 15% is caused specifically by proper tire inflation.

Justification: _____

7. [True/False] In English grammar, 'had better not' must be followed by 'to + infinitive'.

Justification: _____

8. [True/False] The letter 'g' is silent in 'alignment'.

Justification: _____

9. [True/False] The 'School of the Air' was established in Australia to serve urban schools.

Justification: _____

10. [True/False] Under-inflated tires decrease rolling resistance.

Justification: _____

Detailed Explanatory Solutions & Pedagogical Keys

OFFICIAL ANSWER KEY & EXPLANATIONS

- 1. **Correct Answer: A** → 'We should have inflated' correctly expresses regret for an action we failed to do in the past.
- 2. **Correct Answer: B** → The 'h' in 'exhaust' is completely silent (/ɪg'zɔ:st/). 'Acceleration' and 'footprint' have no silent letters.
- 3. **Correct Answer: B** → 'Unless' means 'If ... not'. It already carries a negative meaning, so the following verb must be affirmative to avoid double negation: 'Unless you align your car...'.
4. **Correct Answer: B** → Compound nouns systematically stress the first element: **ECO-driving**.
- 5. **Correct Answer: B** → 'Mitigated' ends in the /ɪd/ sound because the base verb ends in a 't' sound.
- 6. **FALSE:** Aerodynamic drag is caused by air resistance (boosted by cargo roof racks), not tire inflation.
- 7. **FALSE:** 'Had better not' is followed by a bare infinitive (without 'to'). Example: 'You had better not drive fast'.
- 8. **TRUE:** The 'g' is completely silent, phonetic spelling is /ə'laɪn.mənt/.
- 9. **FALSE:** It was established in 1951 to serve remote, geographically isolated children in the Australian Outback.
- 10. **FALSE:** Under-inflated tires increase rolling resistance by 3%, which in turn wastes fuel.

Synthesis Memo 1.1: Expressing Advice & Regrets Grammar Matrix

Grammatical Concept	Formula / Construction	Pedagogical Example	BAC Trap to Avoid
Present Advice	Should + Base Verb	You should accelerate smoothly.	Do NOT use 'to' after 'should'.
Strong Present Advice	Had better + Base Verb	You had better inflate tires.	Do NOT say 'had better to'.
Past Regret (Affirmative)	Should have + Past Participle	We should have driven slower.	Check that V3 form is correct.
Past Criticism (Negative)	Shouldn't have + Past Participle	He shouldn't have idled so long.	Must criticize past negligence.

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